Denaturing SDS polyacrylamide gel electrophoresis

In the presence of anionic detergents like sodium dodecyl sulfate (SDS), proteins lose their tertiary structure and acquire a uniform negative charge. Since most secondary structures are also disrupted, the proteins migrate through a gel matrix at a rate proportional to their polypeptide length.

Protein resolution in SDS-PAGE depends on multiple factors, including acrylamide concentration, crosslinker ratio, and the buffer system used. Lower acrylamide or crosslinker concentrations form larger pores that better resolve high molecular weight proteins, while smaller proteins separate better in high-percentage acrylamide gels. To concentrate the sample in a possibly small volume before separation, a stacking gel with larger pores is often poured on top of a finer resolving gel.

Discontinuous buffer systems further enhance resolution by creating sharp voltage gradients. Laemmli's original Tris-HCl/glycine system (Laemmli, 1970), *Alternative A*, remains the most widely used for proteins between 5 kDa and 250 kDa, although the buffer can reach pH values above pH 9.5 during electrophoresis, potentially leading to deamidation or alkylation of the samples. For sensitive applications such as mass spectrometry, Bis-Tris gels with MES or MOPS running buffers (*Alternative B*) provide improved pH stability and are gentler on proteins. These buffers are especially suitable for small and mid-sized proteins, respectively, but tend to perform poorly for proteins above 100 kDa.

Risk assessment

- **Acrylamide is a KNOWN NEUROTOXIN and LIKELY CARCINOGENIC**
- Work with high voltage power sources

▷ Wear gloves, safety glasses, lab coat

- [ ] Collect acrylamide monomer solutions as HAZARDOUS WASTE
- [ ] DO NOT wash into sewer

Reviewed: Jul 17, 2024

Procedures

›› *Casting SDS polyacrylamide gels*

☐ Gel casting apparatus	☐ Gel comb
☐ Gel cassette	☐ Isopropyl alcohol

(1.) Prepare resolving and stacking gel solutions in the order of the recipes below, but omit ammonium persulfate and TEMED. Invert the solution several times to mix the ingredients. Avoid foaming!

> ***Hint:*** Gel percentage determines how far and how sharply proteins migrate. There is a reference chart at the end of this protocol for approximate resolution ranges. 12% is a good starting point for most proteins in the 20–100 kDa range.

> ***This is why:*** The crosslinker ratio describes the proportion of acrylamide monomer to the crosslinker bis-acrylamide. A ratio of 37.5:1 (2.6% crosslinker) is the most common choice and suitable for most applications. Lower ratios such as 19:1 (5% crosslinker) produce a tighter mesh that improves resolution of large proteins but makes gels more brittle. Higher ratios such as 100:1 create softer gels with larger pores.

> ***This is why:*** Ammonium persulfate (APS) initiates the polymerization of the acrylamide/bis-acrylamide mixture catalyzed by TEMED. Solutions without APS and TEMED can be stored for several months. If TEMED is present, store the premixed solution preferably at a cold place in the dark.

(2.) *Optional:* Filter the solutions through a 0.45 µm pore.

> ***This is why:*** Filtration removes keratin, a common contaminant in mass spectrometry.

(3.) *Optional:* Degas under vacuum for 15 min at room temperature.

> ***Quality assurance:*** For reproducible polymerization, the dissolved oxygen must be removed.

(4.) Assemble the gel cassette. Mark the front plate about 1 cm below the bottom of the comb to indicate the top of the resolving gel. Remove the comb.

(5.) Add ammonium persulfate and TEMED to the resolving gel solution, invert gently several times to mix. Work quickly since polymerization begins immediately.

Note: TEMED aliquots should be replaced every three months, as they slowly oxidize.

(6.) Pour the resolving gel solution into the assembled gel cassette with the comb removed.

Gel fraction	Spacer	Mini Gel (7.2 cm × 8.6 cm)	Midi Gel (13.3 cm × 8.7 cm)
Resolving gel	1.0 mm	5.0 mL	9.0 mL
Stacking gel	1.0 mm	1.5 mL	2.5 mL

(7.) Gently overlay the resolving gel with a few drops of isopropyl alcohol to create a smooth interface.

(8.) Allow 45–90 min for the gel to polymerize. This ensures reproducible pore size. 60 min

Quality assurance: A sharp interface between the gel and the alcohol overlay indicates complete polymerization.

(9.) Pour off the alcohol, rinse with water, dry the area above the resolving gel by touching a strip of filter paper to the edge of the cassette.

(10.) Add ammonium persulfate and TEMED to the stacking gel solution. Invert gently to mix.

(11.) Pour the stacking gel solution in the gel cassette. To avoid spilling, place the pipet tip at an inclined angle and dispense slowly. Fill to the rim before inserting the comb.

(12.) Store gels flat at 4 °C with the comb inserted. Wrap in a wet paper towel and seal in a plastic bag.

Critical: Run handcast gels with discontinuous buffer systems right after casting since the discontinuity gradually disappears. Basic Tris-Cl SDS-PAGE gels must be used within a couple of weeks; Bis-Tris gels are stable for up to six months.

›› *Running SDS polyacrylamide gels*

- ☐ Electrophoresis tank
- ☐ Power supply
- ☐ 20% Iodoacetamide, aqueous
- ☐ Low molecular weight marker R0110
  - *or*: Precision Plus Protein™ Dual Color Standards, Bio-Rad
- ☐ High molecular weight marker R0111
  - *or*: HiMark™ Pre-Stained Protein Standard, Invitrogen

(1.) Remove comb by pulling straight up, slowly and gently. Rinse wells carefully with water.

(2.) Assemble the electrophoresis cell.

(3.) Add fresh running buffer to the reservoir that connects to the stacking gel. Fill the other reservoir with fresh or used running buffer.

Critical: Always fill both inner and outer chambers with the recommended volume of running buffer. Incomplete filling may cause uneven band migration or excessive heating.

Hint: The stacking gel chamber must always be filled with fresh buffer. The buffer in the other chamber may be reused once or twice, especially for short runs. Avoid reuse if previous samples were high in guanidinium, or reducing agent, or if gels are destined for downstream mass spectrometry. Discoloration of the tracking dye during the run may indicate excessive reuse.

(4.) Prepare samples in 1 × sample buffer. Load 0.1–5 µg of purified protein or up to 25 µg crude extract per lane.

Hint: Concentrate samples by lyophilization, spin concentration, gel filtration, or dialysis against polyethylene glycol. To remove potassium, guanidine hydrochloride, or ionic detergents, precipitate with trichloroacetic acid or acetone.

Quality assurance: Ensure an excess of SDS; typically 1.5–3.0 µg SDS per microgram protein. Diluted sample buffer provides approximately 10 µg/µL SDS, which is sufficient for most applications up to 5 µg protein per lane.

(5.) Immediately heat to 80–90 °C for 2 min (5 min for pellets) to denature the proteins.

(6.) *Optional:* Bring to 60 °C, add iodoacetamide to 2% (w/v). Incubate for 30 min at room temperature.

This is why: Iodoacetamide caps free sulfides, preventing back-folding and protein aggregation, which leads to sharper bands and less artifacts such as double bands, blurred zones, or horizontal lines across the gel from excess reductant.

(7.) Remove insoluble material at 17 000 × g for 2 min.

> ***Hint:*** Certain proteins like histones and membrane proteins may not completely dissolve by heating in SDS sample buffer. Add 6–8 M urea or a nonionic detergent such as Triton™ X-100.

> ***Quality assurance:*** Cleared SDS-PAGE samples stored at 4 °C overnight or frozen for longer periods, should be briefly warmed at 37 °C to redissolve SDS. Remove insoluble material by centrifugation.

(8.) Load the appropriate volume of each sample per well. For midi gels, wells typically accommodate 15–40 µL. Use a long pipette tip for loading. Viscous samples often fail to settle into the well and can cause smears.

(9.) Close the lid, connect tank and power supply, perform electrophoresis at constant voltage. 30–60 min

- Run at 5–10 V per centimeter of gel (70–80 V for Mini and Midi gels) for about 10 min until the sample is concentrated at the starting point of the resolving gel.
- Increase the voltage for the resolving phase as recommended. Do not exceed the limits set by the manufacturer for the electrophoresis system.

(10.) Turn the power supply off and disconnect the electrical leads.

(11.) Pop open the gel cassette. Gently float the gel off the plate into a water basin for staining or transfer.

> ***Hint:*** Wet gloves before handling the gel to minimize sticking and tearing. Lift one edge with a dampened plastic wedge.

A › *Tris-Cl/Tris-glycine SDS PAGE (Laemmli)*

- ☐ 0.5 M Tris hydrochloride, pH 6.8 R0102
- ☐ 1.5 M Tris hydrochloride, pH 8.8 R0103
- ☐ 4 × Laemmli sample buffer, pH 6.8 R0104
- ☐ 5 × Tris-Gly running buffer R0105

(1.) Prepare the gel solutions.

- Combine the ingredients below to prepare 24 mL resolving gel solution sufficient for two Midi gels (9 mL each) or four Mini gels (5 mL each) with 1.0 mm spacer. Select the column matching the desired acrylamide percentage.

  With 30% acrylamide-bis acrylamide solutions (other ingredients as below):

Ingredient	Stock	Final	7.0%	8.0%	10.0%	12.0%	15.0%	16.0%
Water, reagent-grade			12.2 mL	11.4 mL	9.8 mL	8.2 mL	5.8 mL	5.0 mL
Acrylamide/bis-acrylamide (37.5:1)	30%		5.6 mL	6.4 mL	8.0 mL	9.6 mL	12.0 mL	12.8 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	Stock	Final	7.0%	8.0%	10.0%	12.0%	15.0%	16.0%
Water, reagent-grade			13.6 mL	13.0 mL	11.8 mL	10.6 mL	8.8 mL	8.2 mL
Acrylamide/bis-acrylamide (37.5:1)	40%		4.2 mL	4.8 mL	6.0 mL	7.2 mL	9.0 mL	9.6 mL
Tris-Cl, pH 8.8	1.5 M	375 mM	6.0 mL	6.0 mL	6.0 mL	6.0 mL	6.0 mL	6.0 mL
SDS	20%	0.1%	120 µL	120 µL	120 µL	120 µL	120 µL	120 µL
TEMED	100%	0.1%	24 µL	24 µL	24 µL	24 µL	24 µL	24 µL
Ammonium persulfate (APS)	10%	0.1%	240 µL	240 µL	240 µL	240 µL	240 µL	240 µL

- For 8 mL stacking gel solution.

  With 30% acrylamide-bis acrylamide solutions (others below):

Ingredient	Stock	Final	4.0%	5.0%
Water, reagent-grade			4.8 mL	4.6 mL
Acrylamide/bis-acrylamide (37.5:1)	30%		1.1 mL	1.3 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	Stock	Final	4.0%	5.0%
Water, reagent-grade			5.1 mL	4.9 mL
Acrylamide/bis-acrylamide (37.5:1)	40%		0.8 mL	1.0 mL
Tris-Cl, pH 6.8	0.5 M	125 mM	2.0 mL	2.0 mL
SDS	20%	0.1%	40 µL	40 µL
TEMED	100%	0.1%	8 µL	8 µL
Ammonium persulfate (APS)	10%	0.1%	80 µL	80 µL

(2.) Dilute samples with 4 × Laemmli sample buffer.

(3.) Run gels in 1 × Tris-Gly running buffer at 180–200 V for 40–60 min.

B › *Bis-Tris/MES and Bis-Tris/MOPS SDS PAGE*

- ☐ 3.5 × Bis-Tris, pH 6.8 R0106
- ☐ 4 × LDS sample buffer, pH 8.5 R0107
- ☐ 20 × MES running buffer, pH 7.3 R0108
- ☐ 20 × MOPS running buffer, pH 7.7 R0109
- ☐ 1 M Sodium bisulfite

(1.) Prepare the gel solutions.

- Combine the ingredients below to prepare 24 mL resolving gel solution sufficient for two Midi gels (9 mL each) or four Mini gels (5 mL each) with 1.0 mm spacer. Select the column matching the desired acrylamide percentage.

  With 30% acrylamide-bis acrylamide solutions (other ingredients as below):

Ingredient	Stock	Final	7.0%	8.0%	10.0%	12.0%	15.0%	16.0%
Water, reagent-grade			11.4 mL	10.6 mL	9.0 mL	7.4 mL	5.0 mL	4.2 mL
Acrylamide/bis-acrylamide (37.5:1)	30%	(*var.*)	5.6 mL	6.4 mL	8.0 mL	9.6 mL	12.0 mL	12.8 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	Stock	Final	7.0%	8.0%	10.0%	12.0%	15.0%	16.0%
Water, reagent-grade			12.8 mL	12.2 mL	11.0 mL	9.8 mL	8.0 mL	7.4 mL
Acrylamide/bis-acrylamide (37.5:1)	40%	(*var.*)	4.2 mL	4.8 mL	6.0 mL	7.2 mL	9.0 mL	9.6 mL
Bis-Tris, pH 6.8	3.5 ×	360 mM	6.9 mL	6.9 mL	6.9 mL	6.9 mL	6.9 mL	6.9 mL
TEMED	100%	0.1%	24 µL	24 µL	24 µL	24 µL	24 µL	24 µL
Ammonium persulfate (APS)	10%	0.06%	144 µL	144 µL	144 µL	144 µL	144 µL	144 µL

- For 8 mL stacking gel solution.

  With 30% acrylamide-bis acrylamide solutions (others below):

Ingredient	Stock	Final	4.0%	5.0%
Water, reagent-grade			4.6 mL	4.4 mL
Acrylamide/bis-acrylamide (37.5:1)	30%	(*var.*)	1.1 mL	1.3 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	Stock	Final	4.0%	5.0%
Water, reagent-grade			4.9 mL	4.7 mL
Acrylamide/bis-acrylamide (37.5:1)	40%	(*var.*)	0.8 mL	1.0 mL
Bis-Tris, pH 6.8	3.5 ×	360 mM	2.3 mL	2.3 mL
TEMED	100%	0.1%	8 µL	8 µL
Ammonium persulfate (APS)	10%	0.06%	48 µL	48 µL

(2.) Dilute samples with 4 × Laemmli sample buffer or 4 × LDS sample buffer.

(3.) Run the gel using one of the following conditions: 30–60 min

- ○ To resolve 5–75 kDa proteins, 1 × MES running buffer, 200 V for 35 min, 4 °C.
- ○ To resolve 50–200 kDa proteins, 1 × MOPS running buffer, 200 V for 50 min, room temperature.

> ***Critical:*** Replace running buffer as soon as the pH indicator in the LDS sample buffer turns yellow.

> ***Critical:*** Reducing agents such as 2-mercaptoethanol (BME) and dithiothreitol (DTT) are not ionized in Bis-Tris and therefore do not enter the gel. To prevent protein reoxidation during electrophoresis, add 5 mM bisulfite (final) to the running buffer.

Materials

Casting SDS polyacrylamide gels

Gel casting apparatus

- *e.g.* **Mini-PROTEAN® Tetra Handcast Systems**, Bio-Rad
- *e.g.* **Criterion™ Empty Cassettes for handcasting**, Bio-Rad

Gel cassette, 0.5–1.5 mm spacer

- **Gel cassette**

Gel comb, 12 wells

- **Gel comb**

Isopropyl alcohol

Running SDS polyacrylamide gels

Electrophoresis tank

- *e.g.* **Mini-PROTEAN® Tetra Cell**, Bio-Rad
- *e.g.* **Criterion™ Cell**, Bio-Rad

Power supply

- *e.g.* **PowerPac™ HC**, Bio-Rad

Iodoacetamide, 20%, aqueous

Molecular weight marker (low)

- **Low molecular weight marker** R0110, Store at −20 °C.
- *or* **Precision Plus Protein™ Dual Color Standards**, Bio-Rad

Molecular weight marker (high)

- **High molecular weight marker** R0111, Store at −20 °C.
- *or* **HiMark™ Pre-Stained Protein Standard**, Invitrogen

Tris-Cl/Tris-glycine SDS PAGE (Laemmli)

Tris hydrochloride R0102, 0.5 M, pH 6.8, Store at 4 °C.

Tris hydrochloride R0103, 1.5 M, pH 8.8, Store at 4 °C.

Laemmli sample buffer R0104, 4 ×, pH 6.8, Store at −20 °C.

Tris-Gly running buffer R0105, 5 ×, Store at room temperature.

Bis-Tris/MES and Bis-Tris/MOPS SDS PAGE

Bis-Tris R0106, 3.5 ×, pH 6.8, Store at 4 °C.

LDS sample buffer R0107, 4 ×, pH 8.5, Stable for 6 months at room temperature.

MES running buffer R0108, 20 ×, pH 7.3, Stable for 6 months at 4 °C.

MOPS running buffer R0109, 20 ×, pH 7.7, Store at −20 °C. Stable for 6 months at 4 °C.

Sodium bisulfite, 1 M

Recipes

Low molecular weight marker (LMW)

Amount	Ingredient		Stock	Final
1.25 mL	Sample buffer		4 ×	1 ×
1 mg	Lysozyme, hen egg white Respiratory sensitization (H334)	[12650-88-3]	14 kDa	0.2 g/L
1 mg	Trypsin inhibitor, soybean Respiratory sensitization (H334)	[9035-81-8]	22 kDa	0.2 g/L
1 mg	Carbonic anhydrase, bovine erythrocytes [9001-03-0] Respiratory sensitization (H334)		29 kDa	0.2 g/L
2 mg	Peroxidase, horseradish Respiratory sensitization (H334)	[9003-99-0]	40 kDa	0.4 g/L
1 mg	Albumin, bovine serum	[9048-46-8]	66 kDa	0.2 g/L
2 mg	Lipoxidase, soybean Respiratory sensitization (H334)	[9029-60-1]	96 kDa	0.4 g/L
To 5 mL	Water, reagent-grade			

Dilute each components in 500 µL 1 × sample buffer. Dispense into 20 µL aliquots. Store at −20 °C. ***Note:*** Check the integrity of each component on a gel before combining. Apply 20 µL low molecular weight marker (LMW) per lane.

Low molecular weight marker (LMW)

1 × Sample buffer, 0.2 g/L Lysozyme, 0.2 g/L Trypsin inhibitor, 0.2 g/L Carbonic anhydrase, 0.4 g/L Peroxidase, 0.2 g/L Albumin, 0.4 g/L Lipoxidase

Composition not fully characterised; assess before use.

☐ Hold for EHS review before disposal

Date: Sign: R0110

High molecular weight marker (HMW)

Amount	Ingredient		Stock	Final
1.25 mL	Sample buffer		4 ×	1 ×
2 mg	Peroxidase, horseradish Respiratory sensitization (H334)	[9003-99-0]	40 kDa	0.4 g/L
1 mg	Albumin, bovine serum	[9048-46-8]	66 kDa	0.2 g/L
2 mg	Lipoxidase, soybean Respiratory sensitization (H334)	[9029-60-1]	96 kDa	0.4 g/L
2 mg	β-Galactosidase, *E. coli*	[9031-11-2]	116 kDa	0.4 g/L
2 mg	Myosin, heavy chain, rabbit muscle [1638095-42-7]		205 kDa	0.4 g/L
To 5 mL	Water, reagent-grade			

Dilute each component in 500 µL 1 × sample buffer. Dispense into 20 µL aliquots. Store at −20 °C. ***Note:*** Check the integrity of each component on a gel before combining. Apply 20 µL high molecular weight marker (HMW) per lane.

High molecular weight marker (HMW)

1 × Sample buffer, 0.4 g/L Peroxidase, 0.2 g/L Albumin, 0.4 g/L Lipoxidase, 0.4 g/L β-Galactosidase, 0.4 g/L Myosin

Composition not fully characterised; assess before use.

☐ Hold for EHS review before disposal

Date: Sign: R0111

Tris hydrochloride (Tris-Cl), pH 6.8, 0.5 M

Amount	Ingredient		Stock	Final
6.0 g	Tris base	[77-86-1]	121.14 g/mol	0.5 M
5 mL	HCl, 37% Skin corrosion (H314)	[7647-01-0]	11.65 M	
To 100 mL	Water, reagent-grade			

Store at 4 °C.

0.5 M Tris hydrochloride (Tris-Cl)

pH 6.8

No GHS hazards at the indicated concentrations

☐ Dilute below the discharge limit

☐ Flush to drain with excess water

Date: Sign: R0102

Tris hydrochloride (Tris-Cl), pH 8.8, 1.5 M

Amount	Ingredient		Stock	Final
45.4 g	Tris base	[77-86-1]	121.14 g/mol	1.5 M
5 mL	HCl, 37%	[7647-01-0]	11.65 M	
	Skin corrosion (H314)			
To 250 mL	Water, reagent-grade			

Store at 4 °C.

1.5 M Tris hydrochloride (Tris-Cl)

pH 8.8

No GHS hazards at the indicated concentrations

- ☐ Dilute below the discharge limit
- ☐ Flush to drain with excess water

Date: Sign: R0103

Laemmli sample buffer, pH 6.8, 4 ×

Amount	Ingredient		Stock	Final
5.0 mL	Tris hydrochloride (Tris-Cl), pH 6.8 R0102		0.5 M	250 mM
0.80 g	SDS	[141-21-3]	288.38 g/mol	8%
	Flammable solid (H228); Serious eye damage (H318)			
2.0 mg	Bromophenol blue	[115-39-9]	670.00 g/mol	0.2 g/L
4.0 mL	Glycerol, anhydrous	[56-81-5]	92.09 g/mol	40%
0.8 mL	2-Mercaptoethanol	☐ [60-24-2]	78.14 g/mol	1.14 M
	Acute dermal toxicity (H310); Serious eye damage (H318); Reproductive toxicity (H361); Organ toxicity, repeated exposure (H373)			

Add reducing agent freshly; dispense into 920 µL aliquots and make up to 1 mL with 80 µL 2-mercaptoethanol just before use. Alternatively, make up with water and add dithiothreitol to 100 mM when diluting samples. Store at −20 °C. ***Note:*** Bromophenol blue is saturated at 1.0 g/L. The high chloride concentration in this sample buffer aids stacking.

4 × Laemmli sample buffer

63 mM Tris-Cl, 2% SDS, 50 mg/L Bromophenol blue, 10% Glycerol, ☐ 285 mM 2-Mercaptoethanol, pH 6.8

DANGER

Acute dermal toxicity; Serious eye damage; Skin sensitization; Organ damage through prolonged or repeated exposure; Reproductive toxicity; Skin irritation; Acute oral toxicity; Aquatic toxicity (long-term)

- ☐ Collect as HAZARDOUS WASTE
- ☐ DO NOT wash into sewer

Date: Sign: R0104

Tris-Gly running buffer, 5 ×

Amount	Ingredient		Stock	Final
15.1 g	Tris base	[77-86-1]	121.14 g/mol	125 mM
72.0 g	Glycine	[56-40-6]	75.07 g/mol	0.96 M
5.0 g	SDS	[141-21-3]	288.38 g/mol	5%
	Flammable solid (H228); Serious eye damage (H318)			
To 1.0 L	Water, reagent-grade			

Expect pH 8.3. Store at room temperature. ***Critical:*** Do not adjust even if above pH 9.0 since glycine (rather than chloride) must remain the trailing ion in the buffer!

5 × Tris-Gly running buffer

25 mM Tris base, 192 mM Glycine, 1% SDS

No GHS hazards at the indicated concentrations

- ☐ Dilute below the discharge limit
- ☐ Flush to drain with excess water

Date: Sign: R0105

Bis-Tris, pH 6.8, 3.5 ×

Amount	Ingredient		Stock	Final
261.6 g	Bis-Tris	[6976-37-0]	209.24 g/mol	1.25 M
	Serious eye damage (H318)			
40 mL	HCl, 37 percent	[7647-01-0]	11.65 M	
	Skin corrosion (H314)			
To 1.0 L	Water, reagent-grade			

Filter sterilize. Store at 4 °C. ***Note:*** This buffer can be prepared at pH 6.5 to aid resolution. Bis-Tris strongly interacts with copper, lead, and other heavy metal cations.

3.5 × Bis-Tris

pH 6.8

DANGER

Serious eye damage

- ☐ Hold for EHS review before disposal

Date: Sign: R0106

LDS sample buffer, pH 8.5, 4 ×

Amount	Ingredient	Stock	Final
1.21 g	Tris base [77-86-1]	121.14 g/mol	1 M
0.80 g	Lithium dodecyl sulfate (LDS) [2044-56-6] Flammable solid (H228); Serious eye damage (H318)	272.40 g/mol	8%
40 µL	EDTA, pH 8.0 R0017	0.5 M	2 mM
0.2 mL	Brilliant Blue G [6104-58-1]	10 g/L	0.2 g/L
0.2 mL	Phenol red [143-74-8]	10 g/L	0.2 g/L
4.0 mL	Glycerol, anhydrous [56-81-5]	92.09 g/mol	40%
To 10 mL	Water, reagent-grade		

Stable for 6 months at room temperature. ***Note:*** 8% LDS can be substituted with up to 4% SDS. Brilliant Blue G gives a sharper dye front in MES and MOPS running buffers and migrates more closely to the moving ion front than bromophenol blue. Phenol red will turn yellow when the running buffer was poorly made or recycled too many times. Commercialized in XT or NuPAGE® systems.

4 × LDS sample buffer

250 mM Tris base, 2% LDS, 500 µM EDTA, 50 mg/L Brilliant Blue G, 50 mg/L Phenol red, 10% Glycerol, pH 8.5

DANGER

Serious eye damage

- ☐ Collect as HAZARDOUS WASTE
- ☐ DO NOT wash into sewer

Date: Sign: R0107

MES running buffer, pH 7.3, 20 ×

Amount	Ingredient	Stock	Final
60.6 g	Tris base [77-86-1]	121.14 g/mol	1 M
97.6 g	2-(*N*-Morpholino)ethanesulfonic acid (MES) [4432-31-9]	195.24 g/mol	1 M
10.0 g	SDS [141-21-3] Flammable solid (H228); Serious eye damage (H318)	288.38 g/mol	2%
20 mL	EDTA, pH 8.0 R0017	0.5 M	20 mM
To 0.5 L	Water, reagent-grade		

expect pH 7.3, do not adjust. Stable for 6 months at 4 °C.

20 × MES running buffer

50 mM Tris base, 50 mM MES, 0.1% SDS, 1 mM EDTA, pH 7.3

WARNING

Skin and serious eye irritation; Respiratory irritation

- ☐ Dilute below the discharge limit
- ☐ Flush to drain with excess water

Date: Sign: R0108

MOPS running buffer, pH 7.7, 20 ×

Amount	Ingredient	Stock	Final
60.6 g	Tris base [77-86-1]	121.14 g/mol	1 M
104.6 g	4-Morpholinepropanesulfonic acid (MOPS) [1132-61-2]	209.27 g/mol	1 M
10.0 g	SDS [141-21-3] Flammable solid (H228); Serious eye damage (H318)	288.38 g/mol	2%
20 mL	EDTA, pH 8.0 R0017	0.5 M	20 mM
To 0.5 L	Water, reagent-grade		

Expect pH 7.7, do not adjust. Dispense into 50 mL aliquots. Store at −20 °C. Stable for 6 months at 4 °C.

20 × MOPS running buffer

50 mM Tris base, 50 mM MOPS, 0.1% SDS, 1 mM EDTA, pH 7.7

WARNING

Skin and serious eye irritation; Respiratory irritation

- ☐ Dilute below the discharge limit
- ☐ Flush to drain with excess water

Date: Sign: R0109

Sodium bisulfite, 1 M

Amount	Ingredient		Stock	Final
10.4 g	Sodium bisulfite	[7631-90-5]	104.06 g/mol	1 M
To 100 mL	Water, reagent-grade			

Prepare freshly before use.

1 M Sodium bisulfite

EPA Safer Choice low-concern

☐ Hold for EHS review before disposal

Date: Sign:

Troubleshooting

Casting SDS polyacrylamide gels

In Step 8:

- Swirls in gel; disturbed protein separation due to unequal or incomplete polymerization of the gel.
  - Ensure alcohol is poured off within one hour to avoid dehydration.
  - For more consistent results, let the resolving gel polymerize overnight at room temperature. Pour off the alcohol overlay within one hour to prevent dehydration of the gel surface.
  - If polymerization is too fast (less than 10 min), reduce APS and TEMED by 25%.
  - If polymerization is too slow (longer than 120 min), increase APS and TEMED concentration by 50%.

In Step 11:

- Air bubbles trapped in the resolving or stacking gel
  - Pour gel solutions slowly along the glass plate. Tilt the cassette slightly to allow bubbles to escape before the gel sets.
  - If bubbles appear after pouring, dislodge by gently tapping the cassette or inserting a thin needle.
- Poor well formation.
  - Degas the monomer solution to remove dissolved oxygen, which inhibits polymerization.
  - Use fresh catalyst stocks and adjust concentration to 0.12% TEMED and 0.06% APS if necessary.
- Webbing; excess acrylamide behind the comb teeth.
  - Use fresh APS and TEMED or slightly increase their concentration to improve polymerization at the interface.

Running SDS polyacrylamide gels

In Step 3:

- Bands are uneven or wonky.
  - Ensure the gel is covered entirely with buffer during running.
  - Check that the gel cassette is properly seated and there are no leaks between chambers.
  - Use a recirculating cooler or run at lower voltage to reduce heating.

In Step 5:

- Protein degradation during sample preparation.
  - Heat samples immediately after adding SDS sample buffer. SDS easily unfolds most proteins while many proteases remain active unless denatured by heating.
  - For labile proteins, particularly if aspartic acid–proline peptide bonds are present, heat the samples for a shorter time or at a lower temperature such as 75 °C for 10 min.
- Unusual band patterns such as doublets or unexpected fast or slow migration
  - Add urea or nonionic detergent to remove residual secondary structure and complete denaturation

In Step 8:

- Extremely viscous samples due to high DNA or RNA content such as crude cell extracts.
  - Treat samples with Benzonase® Nuclease. This recombinant endonuclease lacks proteolytic activity.
  - Vigorously vortex the heated sample or shear the nucleic acids through sonication to reduce viscosity.

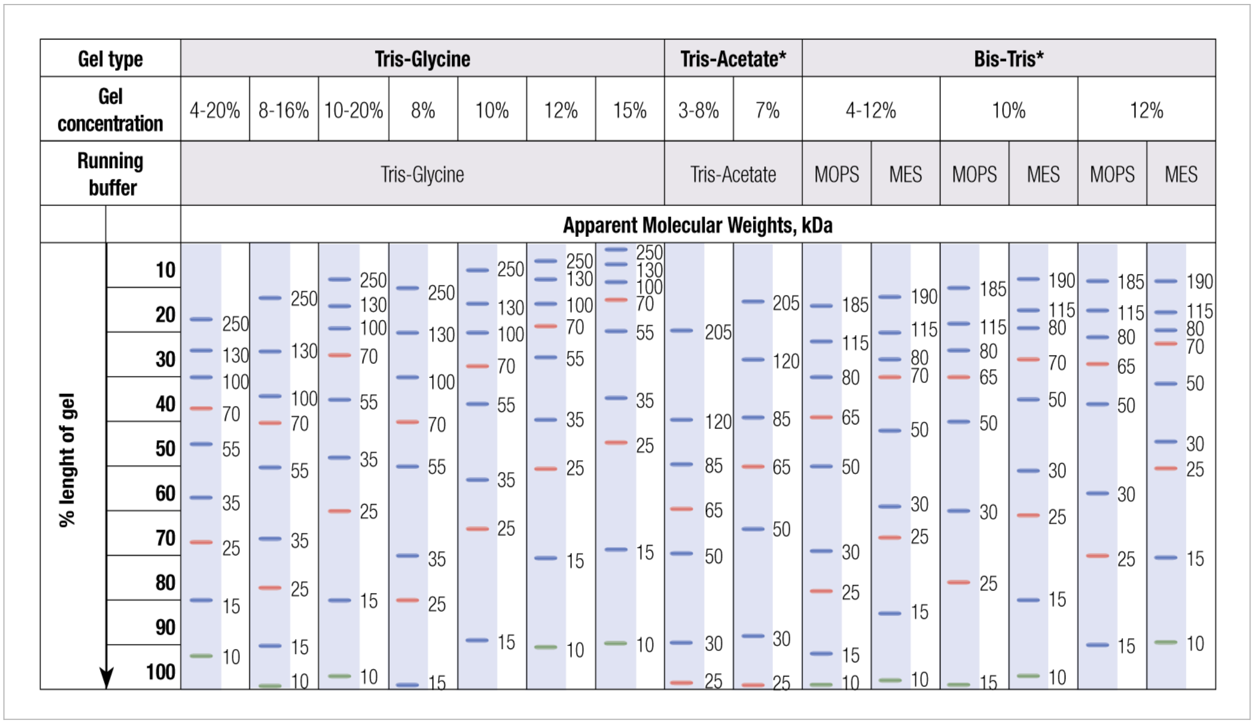
In Step 9:

- Smile effect on bands, distorted lanes, or poor band resolution.
  - Reduce voltage and dissipate heat created during the electrophoresis using a recirculating cooler.

– Frown effect (bands curve downward) at the edges
  - Ensure even gel thickness and uniform polymerization. Check that the gel cassette is level during casting.
  - Verify that the stacking gel has polymerized completely before loading samples.

Resources

Casting SDS polyacrylamide gels



In Step 1: Migration and resolution of denatured proteins in different buffer systems and varying acrylamide concentrations established by a commercial protein ladder. Prestained proteins can exhibit different mobility than the unstained standard. Reproduced from Thermo Scientific Pub. No. MAN0011773.

List of references

U. Laemmli, *Nature* **227**(5259), 680—685 (1970).

Change log

2022-09-20	Benjamin C. Buchmuller	Adaptation as SOP; added SDS/Laemmli and LDS sample buffer recipes. Tris-glycine / Mini-PROTEAN workflow informed by Bio-Rad Mini-PROTEAN and Criterion cell manuals (Bulletin 1658100, Bulletin 6040); Bis-Tris / MES / MOPS pathway drawn from the Tom Muir lab "Pouring and running SDS-PAGE gels" protocol.
2023-01-13	Benjamin C. Buchmuller	Added low and high molecular weight marker recipes from the C. R. Hauck lab "SDS-PAGE gel recipes".
2024-03-11	Benjamin C. Buchmuller	Corrected bromophenol blue content in loading dye concentrate to 0.02% (w/v); literature ranges 0.01–0.50%; less is better.
2024-04-21	Benjamin C. Buchmuller	Corrected diluted molar concentrations of MES and MOPS running buffers to 50 mM; added a section on crosslinker ratios.
2024-07-17	Benjamin C. Buchmuller	Adjusted volumes of resolving gels to accommodate three gels with 1.5 mm spacer or four gels with 1.0 mm.